

PAUL KARONJI WAITHAKA

Full-Stack Software Engineer • Systems Architect • Founder

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Professional Summary

Full-stack software engineer and founder with 3+ years delivering production-grade systems across enterprise, SaaS, mobile, and AI domains. BSc Software Development graduate (KCA University, Dec 2025). Founded WIK Technologies, shipping hardware-licensed POS systems, a property marketplace, AI agents, and consumer PWAs — all serving live users. Deep expertise in offline-first architecture, emerging-market payment integrations (M-Pesa, Paystack), and building complete product ecosystems from zero to production. Comfortable from Electron desktop to React Native mobile to multi-tenant Laravel SaaS.

Technical Skills

Languages	TypeScript, JavaScript (ES6+), PHP, Python, Kotlin, HTML5, CSS3, SQL, Bash
Frontend	React, Next.js, React Native (Expo), Electron, Vite, Tailwind CSS, Radix UI, shadcn/ui, TanStack Query, React Hook Form, Framer Motion, PWA
Backend	Node.js, Express, Laravel 10/12, Next.js API Routes, REST APIs, Socket.IO, WebSockets
Databases	PostgreSQL, MySQL, SQLite, Supabase, Firebase Firestore, Prisma ORM, Eloquent ORM
Cloud & DevOps	Vercel, Cloudflare R2, Firebase, Docker, GitHub Actions CI/CD, Vercel Cron, Upstash Redis, TiDB Serverless
Payments & APIs	M-Pesa Daraja (STK Push & C2B), Paystack, Google Calendar API v3, Africa's Talking, Cloudinary, OAuth2, Resend, VAPID Push, MCP, PHPMailer
Security	JWT, Hardware fingerprinting, RLS, CSRF, Helmet, Rate limiting, Webhook verification, CSP headers, Zod validation, HMAC-SHA256 signing, CVE patching
Architecture	Offline-first, multi-tenant SaaS, hardware licensing, monorepo, Observer pattern, event-driven
Testing & Docs	Jest (45+ tests), Swagger/OpenAPI, CI/CD security invariant testing, 400KB+ README documentation
Tools	Git/GitHub, Docker, Figma, Postman, VS Code

Professional Experience

Oct 2025 – Present Founder & Lead Engineer — WIK Technologies

Nairobi, Kenya

- Founded custom software company delivering web, mobile, cloud, and AI solutions for clients across Kenya and globally.
- Architected and shipped enterprise POS suite (WIK-POS Premium & Standard) — Electron + React + TypeScript, SQLite offline DB, hardware-bound JWT licensing, WhatsApp receipts, customer loyalty CRM, and reorder logistics.
- Built and operate **Pos-License-Server**: centralized licensing authority with hardware fingerprinting (CPU + MAC), 3-tier model (lifetime / 30-day subscription / 5-day trial), and remote revocation.
- Developed **DueSync** (duesync.wikitechnologies.com) — Next.js 15 task-management SaaS with two-way Google Calendar sync, Pomodoro mode, timezone-aware push/email notifications, and Vercel Cron jobs.
- Shipped **K-Track** (ktrack.vercel.app) — commission & payment management SaaS with Paystack M-Pesa + card, USD→KES conversion, Cloudflare R2 file storage, real-time Socket.IO chat, and anti-tampering nonce validation.
- Managed full product lifecycle: requirements gathering, system architecture, development, CI/CD deployment, and client relationships.

Sep 2025 – Dec 2025 Technical Lead (Industrial Attachment) — JHUB Africa

Nairobi, Kenya

- Led development team and architected the **JHUB Africa Innovation Tracker** (innovation.jhubafrica.com) — full-stack project management platform for JKUAT Hub serving innovators, mentors, investors, and admins across Kenya.
- Built in PHP, MySQL, JavaScript, and Docker (Dockerized for production with docker-compose); engineered 4 user roles, 6-stage project lifecycle tracking, mentor assignment engine, threaded comments, document management, resource library, automated email notifications, and activity audit log.
- Conducted code reviews, established team coding standards (PSR-12), and delivered the system on schedule — platform is live in active use.

2023 – Present Freelance Software Developer — Independent

Nairobi, Kenya (Remote)

- Delivered 10+ end-to-end web applications for diverse clients — from e-commerce and property management to agricultural advisory and hospital systems.
- Maintained 98% client satisfaction through clear communication, detailed documentation, and production-quality code with 90+ PageSpeed scores.
- Handled end-to-end engineering from requirements and architecture through deployment and ongoing maintenance with zero missed deadlines.

Selected Projects

MtaaKeys — Property Marketplace & Rental Management Platform

Stack: Next.js 16, Express 5, TypeScript, PostgreSQL, Prisma, Socket.IO, Paystack, Africa's Talking, Cloudinary, Resend

- Full-stack monorepo (Next.js frontend / Express API / Vite admin portal) combining public property marketplace with complete rental management — 5 user roles (Seeker, Agent, Landlord, Land Agent, Admin) with granular Row-Level Security.
- Dual Paystack subscription systems: tiered agent plans (FREE/STARTER/PROFESSIONAL/PREMIUM) gating building & unit quotas, plus Land Agent tier (up to KES 15,000/month). Pay-per-use listing Boosts and automated node-cron lifecycle jobs for renewals, expiry reminders, and tier downgrades.
- Algorithmic featured-listing curation (logarithmic view/inquiry scoring, recency decay, monopoly caps); real-time Socket.IO messaging; Admin Audit Log with exact payload diffing; Africa's Talking SMS + Resend email notifications.
- Security hardening: JWT auth, CSRF, Helmet, rate limiting on auth routes, Paystack webhook signature verification against raw bytes, strict CORS whitelisting, and automated security invariant tests in GitHub Actions CI/CD.

Hiro — Self-Hosted Multi-Model AI Agent Mesh

Stack: [Node.js](#), [TypeScript](#), [Multi-model LLM mesh](#), [PostgreSQL](#), [Telegram API](#), [WhatsApp Cloud API](#)

- Self-hosted AI agent with multi-model mesh workflows, parallel task execution, and a dynamic self-improving skills system.
- Supports document ingestion (PDF, DOCX), web search, memory-backed conversations, voice input/output, and browser canvas for rich output across dual Telegram and WhatsApp interfaces.

DueSync — Smart Task Management PWA & MCP AI Server

Stack: [Next.js 15](#), [TypeScript](#), [PostgreSQL](#), [Prisma](#), [NextAuth v5](#), [Upstash Redis](#), [Vercel Cron](#), [Google Calendar API v3](#), [Radix UI](#)

- Full-featured task management PWA and native Android app (TWA) with two-way Google Calendar sync (OAuth 2.0, proactive token refresh), recurring tasks, Pomodoro focus mode, timezone-aware reminders, daily digest emails, and VAPID push notifications.
- Security hardened: RLS on all tables, strict CSP headers, CSRF protection, Upstash Redis rate limiting, Zod input validation, HTTP-only session cookies, and proactively patched CVE-2025-55182 / CVE-2025-66478.
- Ships a native **Model Context Protocol (MCP)** server (stdio + Streamable HTTP) exposing 13 tools for AI agent integration — enabling agents to list, create, update, complete tasks, and auto-allocate deep-work blocks.

WIK-POS Enterprise Suite — Point of Sale & Licensing Authority

Stack: [Electron](#), [React](#), [TypeScript](#), [SQLite](#), [JWT](#), [KRA eTIMS API](#), [LAN Sockets](#)

- Hardware-bound licensing (CPU + MAC fingerprint), 30-day offline grace periods, customer loyalty CRM, WhatsApp e-receipts, and reorder logistics.
- Modular desktop editions for Retail Supermarkets, Hotel F&B (table management & KOT), and Pharmacy (PPB batch tracking & 60-day FEFO quarantine). Extensively documented (371 KB README); deployed to live retail clients.

Brixton Makunga HMS — Enterprise Hospital Management System

Stack: [Laravel 12](#), [React 19](#), [Vite](#), [MySQL 8](#), [Observer Pattern](#)

- 200+ endpoint RESTful API, 56 database migrations, 35 Eloquent models, automated billing via Observer pattern, real-time triage queue management, and pharmacy FEFO batch expiry registers.
- Multi-role clinical staff access: Doctors, Nurses, Pharmacists, Lab Technicians, Receptionists. Live demo at hmsdemo.wiktechnologies.com.

wiktrack — M-Pesa Automated Income Tracker

Stack: [React Native](#), [Expo 57](#), [Kotlin](#), [Supabase](#), [Firebase FCM](#)

- Android app with native SMS reading via custom Kotlin module — parses incoming M-Pesa SMS messages automatically into structured financial ledgers.
- Local-first architecture with Expo SQLite + background Supabase sync, Firebase FCM push notifications, and Vercel serverless backend.

K-Track — Multi-Tutor Agency & Commission Platform

Stack: [React 19](#), [Vite](#), [Node.js](#), [Express 5](#), [MySQL 8](#) / [TiDB Serverless](#), [Socket.IO](#), [Cloudflare R2](#), [Paystack](#), [Resend](#)

- Full-stack SaaS for commission-based tutor agencies: multi-role access (superadmin / tutor / client), task pooling and claiming, real-time chat, Cloudflare R2 file delivery, quoting, and Paystack payments with USD→KES conversion.
- Security-hardened auth: HMAC-SHA256 timing-safe guest checkout URLs (crypto.timingSafeEqual preventing side-channel attacks), HttpOnly refresh cookies, and scoped file authorization.
- Idempotent boot-time database patch service compatible with TiDB Serverless — auto-migrates schema and adds missing tables on startup.

JHUB Africa Innovation Tracker

Stack: [PHP](#), [MySQL](#), [JavaScript](#), [Docker](#), [PHPMailer](#), [SendGrid](#)

- Full-stack project management platform for JKUAT Hub — 4 user roles, 6-stage project lifecycle, complete application workflow with approve/reject, and mentor assignment engine. Dockerized for production with docker-compose.

Unsent — Private Offline Diary PWA

Stack: [React](#), [Vite](#), [VitePWA](#), [Supabase](#), [IndexedDB](#)

- Fully offline-capable PWA: IndexedDB caching, background sync, Supabase RLS, local password hashing, auto-lock idle timer, voice dictation (Web Speech API), media attachments, and full ZIP data export.

Education

Sep 2021 – Dec 2025 BSc Software Development — KCA University

Nairobi, Kenya

Relevant Coursework: Advanced Web Development • Data Structures & Algorithms • Database Management • Network Security • Software Engineering • System Integration • Mobile Application Development • OOP • Agile Project Management

Certifications

- **Cybersecurity Fundamentals** — IBM SkillsBuild (2025)
- **Data Protection Basics** — Atingi eLearning Platform (Sep 2025)
- **Zoho Creator Developer Certification** — Zoho (Nov/Dec 2025)